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Chapter 2

Numbers

2.1 Natural numbers

The first numbers we ever meet are the whole numbers $0, 1, 2, 3, 4, 5, 6, 7, \dots$ and the natural numbers $1, 2, 3, 4, 5, 6, 7, 8, \dots$ which are written down using the numerals. Thus, we denote the set of natural numbers and whole numbers respectively, as $\mathbb{N}$ and $\mathbb{W}$, and are defined as follows

$$\begin{aligned}\mathbb{N} &= \{1, 2, 3, 4, 5, 6, 7, 8, \dots\} \\ \mathbb{W} &= \{0, 1, 2, 3, 4, 5, 6, 7, 8, \dots\}\end{aligned}$$

The natural numbers can be represented by equally spaced points on a horizontal line where 0 is taken to be the first number.

Natural numbers are ordered. That is; they progress from small to large. On a line, numbers to the left of a given number are less than ($<$) the given number, and numbers to the right are greater ($>$) than the given number. For example, $3 < 5$ while $8 > 5$.

2.1.1 Integers

If the straight line displaying the natural numbers is extended to the left, we get equally spaced points to the left of zero.

These points represent negative numbers which are written as the natural numbers preceded by a minus sign such as -4 . The positive and

negative natural numbers including 0 are collectively called *integers*, denoted by $\mathbb{Z}$. Here, the concept of order still plays a significant role. For example, $-5 < -4$ and $-2 > -4$, this is because -5 appears to the left of -4 while -2 appears to the right of -4 .

It is important so far to note that $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z}$.

2.1.2 Real numbers

Real numbers are numbers that can be expressed as decimals, such as

$$4.00\dots, \frac{-3}{4} = -0.7500\dots, \frac{1}{3} = 0.3333\dots, \text{ and } \sqrt{2} = 1.4142\dots$$

Again, the dots ($\dots$) in each case indicate that the sequence of decimal digits goes on forever. As before, the real numbers can be represented geometrically as points on a (straight) number line called the real line. The symbol $\mathbb{R}$ denotes the set of real numbers.

2.1.3 Rational numbers

Definition 2.1.4. Rational number are numbers that can be expressed in the form of $\frac{a}{b}$, where a and b are integers and $b \neq 0$.

We denote by $\mathbb{Q}$ the set of rational numbers.

Examples of rational numbers are $\frac{1}{3}, \frac{4}{9}, \frac{-4}{9}, \frac{200}{13}$ and $57 = \frac{57}{1}$. Rational numbers are precisely the real numbers with decimal expansions that are either:

1. Terminating (ending in an infinite string of zeroes), for example,

$$\frac{3}{4} = 0.75000\dots = 0.75$$

or

2. Eventually repeating (ending with a block of digits that repeats over and over), for example

$$\frac{23}{11} = 2.090909\dots = 2.\overline{09} \quad (\text{the bar indicates the block of repeating digits})$$

Real numbers that are not rational numbers are called *irrational* numbers. They are characterized by having non-terminating and non-repeating decimal expansions. Examples are

$$\pi, \sqrt{2}, \sqrt[3]{5} \text{ and } \log_{10} 3.$$

Since every decimal expansion represents a real number, there are infinitely many irrational numbers.

Again, we note so far that $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.

Example 2.1.5. Show that each of the following numbers are rational.

1. 1.25

2. $0.\overline{3}$

3. $0.3\overline{59}$.

We recall that an irrational number is a number which cannot be expressed in the form $\frac{a}{b}$, where a and b are integers, $b \neq 0$ and a and b have no common factors.

Example 2.1.6. Prove that $\sqrt{2}$ is an irrational number.

Solution We prove by contradiction. Assume that $\sqrt{2}$ is a rational number. Then there are integers a and $b \neq 0$ with no common factors such that

$$\sqrt{2} = \frac{a}{b}.$$

Squaring both sides yields

$$2 = \frac{a^2}{b^2},$$

so that $a^2 = 2b^2$. This means that a^2 is even, so that a is also even. Thus a has a factor 2. Consequently

$$a = 2p, \quad p \in \mathbb{Z}.$$

From $a^2 = 2b^2$ and

$$a = 2p,$$

we get

$$b^2 = 2p^2,$$

which is even. Again we get that b is also even. This means that both a and b have a common factor 2, which contradicts our earlier assumption. Hence,

$$\sqrt{2} \neq \frac{a}{b}.$$

Therefore it is irrational.

2.1.7 Interval notation

The graph of $\{x : x > 2, x \in \mathbb{R}\}$ is a number line shown below. The open/ unshaded circle at 2 indicates that 2 is not included in the set. Rather than write this set of real numbers using set-builder notation, we can write the set in interval notation as $(2, \infty)$.

In general, the interval notations are:

1. (a, b) represents all the real numbers between a and b , not including a and b . This is an open interval. In set builder notation, we write $\{x : a < x < b, x \in \mathbb{R}\}$. For example, the graph of $(-4, 2)$ is
2. $[a, b]$ represents all the real numbers between a and b , including a and b . This is a closed interval. In set builder notation, we write $\{x : a \leq x \leq b, x \in \mathbb{R}\}$. For example, the graph of $[0, 4]$ is
3. $(a, b]$ represents all the real numbers between a and b , not including a but including b . This is a half open interval. In set builder notation, we write $\{x : a < x \leq b, x \in \mathbb{R}\}$. For example, the graph of $(-1, 3]$ is
4. $[a, b)$ represents all the real numbers between a and b , including a but not b . This is a half open interval. In set builder notation, we write $\{x : a \leq x < b, x \in \mathbb{R}\}$. For example, the graph of $[-4, -1)$ is

Subsets of the real numbers whose graphs extend forever in one or

both directions can be represented by interval notation using the infinity symbol ∞ or the negative infinity symbol $-\infty$. They are:

1. $(-\infty, a)$
2. (b, ∞)
3. $(-\infty, a]$
4. $[b, \infty)$
5. $(-\infty, \infty)$

Example 2.1.8. Let $U = [0, 10)$ and define $A = [0, 5]$, $B = [2, 8]$ and $C = (1, 6)$. Find each of the following sets and display them on the number line.

1. $A \cap C$
2. $A \cap B$
3. $A \cap B^c$
4. $A^c \cap C^c$

Definition 2.1.9. Absolute value of a number

The absolute value of the real number a , denoted by $|a|$, is defined by

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}.$$

Example 2.1.10. Find the absolute value of each of the following:

1. $|5|$
2. $|-4|$
3. $|0|$

Example 2.1.11. Simplify

$$|x - 3| + |x + 2| \text{ given that } -1 \leq x \leq 2.$$

2.1.12 Powers

The arithmetic operation of raising a number to a power is understood to be a repetitive multiplication. For example: $10 \times 10 \times 10 \times 10 = 10^4$; the number 10 multiplied by itself 4 times. In 10^4 , 4 is called the power or index (plural indices), while 10 is called the base.

2.1.13 Laws of powers

Let $a, n, m \in \mathbb{R}$ with $a \neq 0$. We have that

1. $a^0 = 1$

2. $a^{-m} = \frac{1}{a^m}$. For example, $2^{-2} = \frac{1}{2^2} = \frac{1}{4}$.

3. $a^m \times a^n = a^{m+n}$. To multiply exponential expressions with the same base, copy the base and add the powers. For example, $5 \times 5^2 = 5^{1+2} = 5^3 = 125$.

4. $a^m \div a^n = a^{m-n}$. To divide exponential expressions with the same base, copy the base and subtract the indices. For example, $y^9 \div y^5 = y^{9-5} = y^4$.

5. $(a^m)^n = a^{mn}$. Example $(2^3)^2 = 2^6$

6. $a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$.

Example 2.1.14. Simplify each of the following :

1. $(125)^{\frac{2}{3}}$

2.

$$\frac{5x^{-1}y^{-4}}{(3y^5)^{-2}x^9}$$

3.

$$\left(\frac{24a^3b^{-8}}{6a^{-5}b} \right)^{-1}.$$

2.2 Radicals and rational powers

In this section, students will learn how to simplify expressions involving square roots, rationalize denominators and use rational powers/indices.

2.2.1 Square roots

If $b^2 = a$, then b is called a square root of a . The symbol $\sqrt{}$ is used to denote the positive or principal square root of a number, while $-\sqrt{}$ denotes the negative square root.

1. $\sqrt{25} = 5$ because $5^2 = 25$ and 5 is positive.
2. $\sqrt{100} = 10$ because $10^2 = 100$ and 10 is positive.

The symbol $\sqrt{}$ that we use to denote the positive square root is called a *Radical sign*. The number under the radical sign is called the *Radicand*. Together $\sqrt{a}$ is called a *radical expression*.

1. $\sqrt{64}$
2. $\sqrt{9} + \sqrt{16}$

A rational number that is a square of another rational number is called a *perfect square*. All the radicands in the previous example are perfect squares. $\frac{1}{4}$ is a perfect square because $(\frac{1}{2})^2 = \frac{1}{4}$.

Theorem 2.2.2. *Product rule of square roots*

If a and b are positive real numbers, then

$$\sqrt{ab} = \sqrt{a}\sqrt{b}.$$

Example 2.2.3. Use the product rule to simplify

1. $\sqrt{500}$
2. $\sqrt{6x} \times \sqrt{3x}$

Another property for square roots involves division.

Theorem 2.2.4. Quotient rule of square roots

If a and b are positive real numbers, then

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}.$$

Example 2.2.5. Use the quotient rule to simplify

1. $\sqrt{\frac{100}{9}}$
2. $\frac{\sqrt{48x^3}}{\sqrt{6x}}.$

Adding and subtracting square roots

Two or more square roots can be combined using the distributive property provided that they have the same radicands. Such radical expressions are called like radicals. For example, $7\sqrt{11} + 6\sqrt{11} = (7 + 6)\sqrt{11} = 13\sqrt{11}.$

Example 2.2.6. Add or subtract as indicated.

1. $7\sqrt{2} + 5\sqrt{2}$
2. $\sqrt{5x} - 7\sqrt{5x}$
3. $7\sqrt{3} + \sqrt{12}$
4. $4\sqrt{50x} - 6\sqrt{32x}$

2.2.7 Rationalizing the denominator

Rationalizing the denominator is the process that involves rewriting a radical expression as an equivalent expression in which the denominator no longer contains any radical.

If the denominator consists of the square root of a natural number that is not a perfect square, multiply the numerator and the denominator by the smallest radical expression that produces the square root of a perfect square in the denominator.

Example 2.2.8. Rationalize the denominator.

1. $\frac{15}{\sqrt{6}}$

2. $\frac{12}{\sqrt{8}}$

Radical expressions that involve the sum and difference of the same two terms are called *conjugates*. For example, $\sqrt{a} + \sqrt{b}$ and $\sqrt{a} - \sqrt{b}$ are conjugates of each other. What happens when two conjugates are multiplied? Let us see

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = (\sqrt{a})^2 - (\sqrt{b})^2 = a - b$$

To rationalize the denominator if the denominator contains two terms with one or two square roots, we multiply the numerator and the denominator by the conjugate of the denominator.

Example 2.2.9. Rationalize the denominator.

1. $\frac{7}{\sqrt{5} + \sqrt{3}}$

2. $\frac{h}{\sqrt{x+h} - \sqrt{x}}$

Similarly, one can easily rationalize the numerator by considering the discussion above.

2.3 Complex numbers

A complex number is an expression of the form $a + bi$, where $a, b \in \mathbb{R}$, and $i^2 = -1$ or equivalently $\sqrt{-1} = i$. The set of such numbers is called the set of complex numbers and is denoted by $\mathbb{C}$.

If $z = a + bi \in \mathbb{C}$, then we call a the real part of z and b the imaginary part of z . We write this as $a = \text{Re}(z)$ and $b = \text{Im}(z)$. The representation of a complex number in the form $a + bi$ is called the Cartesian form of the complex number. If $b = 0$, then $z = a$, and z is said to be purely real. Similarly, z is purely imaginary if $a = 0$.

Example 2.3.1. Let $z = -3 - 4i$. State $\text{Re}(z)$ and $\text{Im}(z)$.

2.3.2 Equal complex numbers

Definition 2.3.3. Two complex numbers $z_1 = a + bi$ and $z_2 = c + di$ are equal ($z_1 = z_2$) if and only if $a = c$ and $b = d$.

Example 2.3.4. 1. State the value of x and of y given that $x + yi = 5 + 4i$.

2. Find the value of a and b if $(a + b) + (a - b)i = 7 + 2i$.

2.3.5 Addition and subtraction of complex numbers

Let $z = a + bi$ and $\omega = c + di$ be two complex numbers.

$$z + \omega = (a + bi) + (c + di).$$

Although the real and the imaginary parts can not be combined, we can remove the brackets and total up like terms. That is

$$\begin{aligned} z + \omega &= a + bi + c + di \\ &= (a + c) + bi + di \\ &= (a + c) + (b + d)i. \end{aligned}$$

Example 2.3.6. Compute $z + \omega$ if $z = 4 + 5i$ and $\omega = 3 - 2i$.

Similarly, if $z = a + bi$ and $\omega = c + di$, then

$$\begin{aligned} z - \omega &= (a + bi) - (c + di) \\ &= a + bi - c - di \\ &= (a - c) + (b - d)i. \end{aligned}$$

Example 2.3.7. Find $z - \omega$ if $z = 4 + 5i$ and $\omega = 3 - 2i$.

2.3.8 Multiplication of complex numbers

Let $z = a + bi$ and $\omega = c + di$ be two complex numbers.

$$z\omega = (a + bi)(c + di).$$

These are multiplied in the same way as you would determine the product $(3x + 4y)(2x + 5y)$. Thus,

$$\begin{aligned} z\omega &= (a + bi)(c + di) \\ &= ac + adi + bi(c) + bdi \\ &= ac + adi + bci - bd \\ &= (ac - bd) + (ad + bc)i. \end{aligned}$$

Example 2.3.9. Given that $z = 3 + 4i$ and $\omega = 2 + 5i$. Compute $z\omega$.

If the expression contains more than two factors, we multiply the factors together in stages.

Example 2.3.10. Multiply $(3 + 4i)(5 - 8i)(1 - 2i)$.

From the previous example, we observed that $i^3 = -i$, because $i^2 = -1$. Continuing, we see that $i^4 = i^2 \times i^2 = -1 \times -1 = 1$, $i^5 = i$ and $i^6 = -1$.

In summary, to calculate any high power of i , you can convert it to a lower power by taking the closest multiple of 4 that is no longer bigger than the exponent and subtract this from the exponent. For example, $i^{99} = i^{(96+3)} = i^3 = -i$.

Example 2.3.11. Simplify each of the following:

1. i^{17}
2. i^{120}
3. i^{64002}

2.3.12 Conjugate complex numbers

Compute $(5 + 8i)(5 - 8i)$.

A pair of complex numbers like these are called *conjugate complex numbers*, and the product of conjugate complex numbers is always purely real. In general if $z = a + bi$, then the conjugate of z , denoted by $\bar{z}$, is given by $\bar{z} = a - bi$.

Example 2.3.13. Given that $z = 5 + 4i$ compute

1. $z - \bar{z}$

2. $z + \bar{z}$

3. $z\bar{z}$

2.3.14 Division of complex numbers

Division of a complex number by a real number is very easy. For example, if $z = 5 - 4i$, then $\frac{z}{3} = \frac{5-4i}{3} = \frac{5}{3} - \frac{4}{3}i$. However, how do we manage with $\frac{7-4i}{4+3i}$?

If we could convert the denominator into a real number, we could divide out as in the example above. This can easily be done by multiplying the denominator by its complex conjugate.

Example 2.3.15. Simplify each of the following:

1.

$$\frac{3 + 2i}{1 - 3i}$$

2.

$$\frac{1 + i}{3 - 2i} + 2 + 4i$$

Example 2.3.16. Solve each of the following for a and b .

1.

$$\frac{3i - 2}{2i - 6} = 3a + 3(b - 1)i.$$

2.

$$\frac{2}{i} - 2b + i^3 = \frac{i + a}{i} + b.$$

2.3.17 Plotting complex numbers

To graph complex numbers, you simply combine the ideas of the real-number coordinate plane and the Gauss or Argand coordinate plane to create the complex coordinate plane.

In other words, given a complex number $a + bi$, you take the real portion of the complex number (a) to represent the x -coordinate, and you take the imaginary portion (b) to represent the y -coordinate. Although you graph complex numbers much like any point in the real-number coordinate plane, complex numbers are not real. The x -coordinate is the only real part of the complex number, so you call the x -axis the real axis and the y -axis the imaginary axis when graphing in the complex coordinate plane.

Example 2.3.18. Plot each of the following on the complex coordinate plane.

1. $4i$

2. $2 + 3i$

3. $-1 - 4i$

4. $-2 + i$

5. $\frac{1}{2} - 3i$.